

COLLECTING DATA & SCOPE OF INFERENCE

new 1.10–1.13 • old 3.1–3.5 • ★ old 3.6

ZONE A • HOW THE DATA ARE COLLECTED

Data from random sample or randomized experiment

SAMPLING METHODS

SRS: every possible sample of size n has the same chance.

Stratified: divide into similar groups; randomly sample within each.

Cluster: randomly choose whole groups; include everyone in them.

Systematic: start randomly, then select every k th individual.

SOURCES OF BIAS

Undercoverage: some groups are left out or hard to reach.

Nonresponse: selected people do not respond.

Response bias: wording or reporting distorts answers.

Voluntary response: people choose to join the sample.

Convenience: choose people who are easiest to reach.

EXPERIMENTS

Control: hold other conditions alike; compare treatments.

Random assignment: use chance to form treatment groups.

Replication: use enough experimental units.

Completely randomized: assign all units at random.

Randomized block: group similar units, then randomize within.

Matched pairs: compare two related measurements or units.

Placebo: an inactive treatment.

Blinding: hide treatment identity from subjects or evaluators.

Confounding: effects are mixed and cannot be separated.

ZONE B • WHAT THE WORDS MEAN

THE STUDY

Observational: record what happens.

Experiment: impose a treatment.

THE PEOPLE OR OBJECTS

Population: the full group of interest.

Sample: the group studied.

THE MEASUREMENTS

Treatment: a condition imposed.

Response variable: the outcome measured.

ZONE C • WHAT CAN WE CONCLUDE?

	Random assignment: YES	Random assignment: NO
Random sample: YES	Generalize: yes Cause and effect: yes	Generalize: yes Cause and effect: no
Random sample: NO	Generalize: no Cause and effect: yes	Generalize: no Cause and effect: no

ZONE D • WATCH OUT + BEYOND

Random **selection** lets you generalize to the population.

Random **assignment** supports cause and effect.

★ BEYOND THE EXAM

Re-randomize groups many times. If the observed difference is rare, it is evidence against chance alone.